

International Journal of Simulation and Process Modelling

ISSN online: 1740-2131 - ISSN print: 1740-2123

<https://www.inderscience.com/ijspm>

Deep learning and multi-agent simulation for spatiotemporal inventory management in e-commerce

Haibo Zhu, Hongmei Wu

DOI: [10.1504/IJSPM.2026.10077092](https://doi.org/10.1504/IJSPM.2026.10077092)

Article History:

Received:	08 December 2025
Last revised:	12 January 2026
Accepted:	15 January 2026
Published online:	27 March 2026

Deep learning and multi-agent simulation for spatiotemporal inventory management in e-commerce

Haibo Zhu*

School of Computer and Information Engineering,
Harbin University of Commerce,
Harbin, 150028, China
Email: hsdjulien@sina.com
*Corresponding author

Hongmei Wu

Agriculture and Rural Affairs Bureau of Songbei District,
Harbin, 150028, China
Email: tupoedu@163.com

Abstract: E-commerce inventory management faces the dual challenge of handling complex spatiotemporal demand variations and coordinating decisions across distributed warehouse networks. To address this, we propose an integrated simulation framework that combines deep learning for demand forecasting with multi-agent reinforcement learning for inventory optimisation. Our approach employs a spatiotemporal graph network to capture demand dependencies and a value-decomposition network for collaborative decision-making. Validated on the real-world M5 dataset, the framework demonstrates significant improvements in both forecast accuracy and operational efficiency compared to traditional and state-of-the-art methods. Notably, it achieves a substantial reduction in total costs while maintaining high service levels. This work contributes a novel, scalable solution for intelligent inventory management, with direct implications for enhancing supply chain resilience in e-commerce.

Keywords: inventory management; deep reinforcement learning; DRL; spatiotemporal forecasting; multi-agent systems; simulation optimisation.

Reference to this paper should be made as follows: Zhu, H. and Wu, H. (2026) 'Deep learning and multi-agent simulation for spatiotemporal inventory management in e-commerce', *Int. J. Simulation and Process Modelling*, Vol. 23, No. 5, pp.11–23.

Biographical notes: Haibo Zhu is an Associate Professor at the School of Computer and Information Engineering, Harbin University of Commerce, China. He obtained his Master's degree from Harbin University of Commerce in 2004, China. His research interests include e-commerce, logistics management, and simulation optimisation.

Hongmei Wu works at the Agriculture and Rural Affairs Bureau of Songbei District, Harbin, China. She obtained her Master's degree from Northeast Agricultural University in 2006, China. Her research interests include agricultural economic management and rural e-commerce.

1 Introduction

The rapid development of e-commerce has reshaped the global retail landscape, presenting businesses with both unprecedented opportunities and challenges to their supply chain management systems to unprecedented complexity and uncertainty. Inventory management, as the pivotal link connecting market demand with physical operations, has long been regarded as a critical trade-off between cost control and service levels for enterprises (Harahap et al., 2025). For example, inventory levels directly determine warehouse space utilisation, the efficiency of order picking operations, the balance of packaging lines, and the

responsiveness and cost of last-mile delivery. Too high inventory will lead to storage congestion and capital occupation, while too low inventory will lead to order delay, split delivery and emergency replenishment, which will significantly drive up logistics costs and affect customer experience. These physical operations prominently include warehousing, order picking, packing, and last-mile delivery logistics, all of which are directly influenced by inventory availability and placement. An efficient inventory system can minimise holding costs – such as capital tied up, storage losses, and product obsolescence – while meeting rapidly changing customer demands. Conversely, flawed decisions directly lead to lost sales and diminished customer loyalty.

Under traditional business models, inventory decisions largely relied on managerial experience and relatively static operations research models, such as the classic (s, S) strategy or the Newsvendor model. However, the uniquely high-dimensional nature of e-commerce environments (tens of thousands of products), multi-tiered structures (central warehouses, regional warehouses, front-end warehouses), and demand patterns subject to sharp fluctuations driven by promotions, holidays, and social media make these traditional methods inadequate in terms of agility, accuracy, and scalability (Chong et al., 2022). For instance, a static (s, S) policy becomes computationally prohibitive to calibrate individually for millions of SKUs across a distributed network, leading to simplistic and uniform parameter application. More intractable is the complex spatiotemporal dependency embedded in demand signals – namely, the dynamic correlations between sales volumes across different regions and product categories – which poses a formidable challenge to any management model (Wang et al., 2024). Specifically, the inability of classical models to handle non-stationary, high-dimensional data is a primary driver for the adoption of deep learning in this domain. Therefore, developing an intelligent inventory management framework capable of adapting to environmental changes, learning complex patterns from massive historical data, and performing forward-looking dynamic optimisation is not only an urgent practical need for enterprises but also a highly challenging and cutting-edge research topic in academia.

To address this challenge, academia and industry have turned their attention to the rapidly evolving field of deep learning, particularly its applications in time series forecasting. In recent years, models such as time-convolutional networks (TCNs), Long short-term memory (LSTM) networks and their variants, as well as attention-based Transformer architectures, have achieved remarkable success across diverse forecasting tasks (Lim and Zohren, 2021). These models can automatically capture long-term dependencies and nonlinear patterns in data, demonstrating immense potential when handling large-scale, high-dimensional sales data compared to traditional autoregressive integrated moving average (ARIMA) or exponential smoothing methods. For instance, in the renowned M5 Forecasting competition, top-performing solutions predominantly employed complex ensembles of deep learning models, demonstrating their superior ability to surpass traditional statistical methods in demand forecasting accuracy (Makridakis et al., 2022). Subsequent research further explored the application of graph neural networks (GNNs) in capturing hierarchical relationships between products, enhancing forecast consistency. However, a widely overlooked critical limitation is that the vast majority of studies treat forecasting as an isolated, purely accuracy-optimisation problem (Zhao et al., 2021). In practical inventory management systems, the ‘most accurate’ forecast does not necessarily lead to the ‘optimal’ inventory decision. Prediction uncertainty, demand correlations between

different commodities, complementary and substitution effects, and the complexity of inventory cost structures all imply a disconnect between the prediction module and the decision-making module. This disconnect constitutes the first major gap between existing data-driven methods and practical application.

Meanwhile, reinforcement learning (RL), particularly Deep reinforcement learning (DRL), offers an exceptionally compelling paradigm for sequential decision-making problems, demonstrating broad application prospects in inventory dynamics optimisation (Jing and Zhang, 2023). Unlike rule-based strategies, DRL agents autonomously learn optimal actions – such as order quantities – under specific conditions (e.g., current inventory levels, inventory in transit) through continuous interaction with the environment. This learning maximises long-term cumulative rewards (i.e., minimising total cost). Pioneering work has applied DRL to diverse inventory problems with encouraging results, demonstrating its potential in handling complex scenarios involving random demand and lost sales (Gijbrecchts et al., 2022). Recent research has further evolved from single-agent models toward the more realistic multi-agent reinforcement learning (MARL) paradigm to simulate coordination and competition among distributed warehouses (Liu et al., 2025). However, most current research remains constrained by overly simplistic assumptions. Models are often constructed for single-warehouse, single-product scenarios or lack effective collaborative learning mechanisms in multi-agent environments (El Shar et al., 2022). Such simplifications severely deviate from the inherently networked and collaborative nature of e-commerce supply chains. Real-world inventory management constitutes a quintessential multi-agent system: geographically dispersed warehouses make decisions based on local observations, while these decisions mutually influence and constrain one another through shared logistics networks and common service objectives (Sobh, 2024). Forcibly simplifying a multi-warehouse, multi-item collaborative optimisation problem into a series of independent single-agent problems ignores the system’s inherent synergies and competitive dynamics, making global optimisation unattainable. This lack of modelling capability represents the primary bottleneck hindering the practical deployment of existing DRL methods.

Furthermore, this study observes a distinct divide within the current research landscape: one branch comprises deep learning methods whose ultimate goal is to enhance prediction accuracy, while the other encompasses RL approaches aimed at learning strategies under given (often simplified) demand distributions. The former lacks extension to downstream decision-making, while the latter often oversimplifies demand models or relies on inaccurate predefined distributions, resulting in learned strategies that perform poorly under real-world fluctuating demand (Castro Moraes et al., 2023). This separation of ‘prediction’ and ‘decision-making’ severely constrains the upper limit of an intelligent system’s overall performance. An ideal

intelligent inventory management system should be an organism where perception and decision-making are tightly coupled (Dey and Seok, 2024). It must leverage advanced perception technologies (deep learning) to make precise, fine-grained inferences about future demand and its uncertainties, while simultaneously employing advanced decision-making technologies to integrate these inferences into globally optimised collaborative strategies. This deep integration requires models to comprehend not only temporal dynamics but also spatial correlations (warehouse networks, product structures). While recent research has begun exploring end-to-end ‘forecast-to-decision’ frameworks, significant theoretical gaps and methodological shortcomings persist in handling the massive, hierarchical spatiotemporal data unique to e-commerce inventory management (Rishab, 2022).

Given this, the present study aims to bridge the aforementioned gap by exploring the synergistic integration of deep learning and MARL in dynamic inventory management for e-commerce. This research seeks to address a core question: How can an integrated, end-to-end simulation framework be constructed to simultaneously leverage the strengths of deep spatiotemporal networks in demand sensing and the potential of MARL in distributed collaborative decision-making, thereby achieving global dynamic optimisation for large-scale e-commerce inventory systems? To validate the proposed framework’s effectiveness, this study will leverage the complex hierarchical M5 dataset derived from real-world e-commerce scenarios to construct a high-fidelity inventory management simulation environment. This approach aims to test the proposed method’s efficacy and superiority under conditions approximating real-world conditions.

Our work addresses key gaps in current approaches, namely the disconnect between prediction and decision-making, the lack of effective multi-agent coordination in inventory systems, and the under-exploration of simulation-based optimisation for spatiotemporal demand. Experimental results on the M5 dataset show that DC-STGN-MARL reduces total operational costs by 13.6% and improves forecast accuracy by 10.5% over strong baselines, while maintaining a 95.8% service level. The main contributions of this paper are threefold:

- 1 a novel integrated simulation framework that tightly couples deep forecasting with MARL
- 2 a decomposition-coordinated spatiotemporal graph network for accurate and uncertainty-aware demand modelling
- 3 empirical validation demonstrating superior performance in cost, service, and efficiency metrics.

2 Related work and theoretical background

2.1 Classic inventory management theories and models

Classical inventory management theories, such as the Newsvendor and (s, S) models, provide foundational principles but face limitations in e-commerce contexts. Key literature on their adaptations and critiques is summarised below: Classic inventory management theory is founded on operations research and probability statistics, with its core objective being to balance holding costs, ordering costs, and stockout costs under uncertain demand. The boyar model serves as the fundamental framework for addressing inventory decisions for single-period products, aiming to determine the optimal order quantity Q^* that minimises expected costs. This optimal solution corresponds to the critical quantile of the cumulative demand distribution function, namely:

$$F(Q^*) = \frac{C_u}{C_u + C_o} \quad (1)$$

where C_u denotes the unit stockout cost, and C_o denotes the unit excess inventory cost. For multi-period scenarios, the (s, S) strategy is one of the most widely applied: when the inventory level I_t falls below the reorder point s , the system places an order to raise the inventory level to the target level S , i.e., the order quantity $Q_t = S - I_t$. The optimisation of this strategy hinges on precisely determining s and S to minimise long-term average costs (Jackson et al., 2020).

Although these models are theoretically elegant, their application in real-world e-commerce environments faces significant challenges. First, they typically assume demand follows specific probability distributions (such as normal or Poisson distributions), whereas e-commerce demand exhibits high non-stationarity and volatility due to promotions, competition, and social media influences, making it difficult to characterise with simple distributions (Zhao, et al., 2021). Second, the computational complexity of classical models escalates dramatically when handling multi-product, multi-tier inventory systems. This often necessitates reliance on heuristic methods for approximate solutions, potentially leading to suboptimal decision outcomes (Chong et al., 2022). These challenges, particularly in modelling non-stationary and high-dimensional demand patterns, have motivated a paradigm shift towards data-driven deep learning methodologies.

2.2 Deep learning-based demand forecasting method

Deep learning offers a new paradigm for capturing complex patterns in demand sequences through its powerful representation learning capabilities. LSTM networks effectively mitigate the vanishing gradient problem via their sophisticated gating mechanisms (input gate i_t , forget gate f_t , output gate o_t , and cell state C_t), enabling them to learn long-term dependencies. Their core computation can be expressed as:

$$h_t = LSTM(x_t, h_{t-1}, C_{t-1}; \Theta) \quad (2)$$

where h_t denotes the hidden state at time step t , x_t represents the input features, and Θ refers to the model parameters (Lim and Zohren, 2021).

In recent years, the Transformer architecture has achieved breakthrough progress in time series forecasting through its global attention mechanism. At its core lies the self-attention layer, which calculates the correlation strength between sequence elements via query (Q), key (K), and value (V) matrices:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

where d_k is the scaling factor (Vaswani et al., 2017). This mechanism enables parallel processing of sequences and captures long-term dependencies far beyond the capabilities of LSTMs. This is primarily due to the self-attention mechanism, which allows the model to weigh the importance of any time step in the past directly when processing the current step, unlike the sequential processing of LSTMs. Transformer-based models, such as the Temporal Fusion Transformer, demonstrate outstanding performance on large-scale prediction tasks like the M5 competition by integrating static covariates, known future inputs, and temporal features (Makridakis et al., 2022).

However, existing forecasting studies share a common flaw: model optimisation typically prioritises prediction accuracy metrics [such as weighted mean absolute percentage error (WMAPE) or root mean square error (RMSE)] as the ultimate goal, while neglecting how forecast outcomes serve downstream inventory decisions. A model excelling in WMAPE metrics may concentrate its prediction errors precisely within inventory decision sensitivity zones, leading to costly stockouts or overstocking. This disconnects between prediction and decision-making represents a significant limitation in current research.

2.3 Applications of RL in inventory management

RL models inventory management as a sequential decision-making process, where an agent learns optimal policies through interaction with the environment. This problem is typically formalised as a Markov decision process (MDP), consisting of a state space $\mathcal{S}$, an action space $\mathcal{A}$, state transition probabilities $\mathcal{P}$, and a reward function $\mathcal{R}$. The agent's objective is to learn a policy $\pi(a | s)$ that maximises the expected cumulative discounted reward $\mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t R_t]$. Where $\gamma \in [0, 1]$ is the discount factor (Thrun and Littman, 2000).

DRL employs deep neural networks as function approximators to handle high-dimensional state spaces. For instance, deep Q-network (DQN) learns the optimal

action-value function $Q^*(s, a)$ by minimising temporal difference error:

$$L(\theta) = \mathbb{E}\left[\left(R + \gamma \max_{a'} Q(S', a'; \theta^-) - Q(S, A; \theta)\right)^2\right] \quad (4)$$

where θ and θ^- represent the parameters of the online network and target network, respectively (Mnih et al., 2015). In inventory management, DRL has been demonstrated to effectively handle issues such as random demand and lost sales, with its performance approaching or even surpassing carefully tuned baseline strategies under certain settings (Gijsbrechts et al., 2022).

Current research is expanding from single-agent models to MARL to address coordination challenges in distributed inventory systems. However, most existing MARL approaches face two major challenges in e-commerce inventory scenarios. First is environmental non-stationarity: simultaneous learning by multiple agents leads to dynamically changing environments perceived by each agent, hindering convergence. Second is the credit allocation problem: how to fairly distribute the system's global reward to the local actions of individual agents (El Shar et al., 2022). For example, if a global cost reduction occurs, it is challenging to determine whether credit is due to one warehouse for avoiding overstocking or to another for preventing a stockout that would have triggered trans-shipment costs. Although algorithms like value decomposition networks (VDN) attempt to address this by decomposing the joint Q-function into the sum of

individual Q-functions ($Q_{tot}(\vec{s}, \vec{a}) \approx \sum_{i=1}^N Q_i(s_i, a_i)$), such

linear assumptions are often overly simplistic in complex nonlinear inventory systems (Liu et al., 2025).

In summary, existing research has achieved significant progress in the distinct domains of deep forecasting and reinforcement decision-making, yet their development remains largely disconnected. Deeply integrating high-precision spatiotemporal forecasting with decision frameworks capable of multi-agent coordination to construct an end-to-end, perception-decision integrated intelligent inventory management system represents a critical scientific challenge that urgently requires resolution in this field. This constitutes the primary research objective of this paper.

3 Methodology: integrated simulation framework

3.1 Problem formulation

This study models the dynamic management of e-commerce inventory as a decentralised partially observable Markov decision process (Dec-POMDP). This model is suitable for collaborative decision-making systems composed of multiple agents, where each agent can only access partial observational information.

Consider a system composed of N agents (representing N warehouses or product categories) interacting at discrete

time steps t . A Dec-POMDP can be defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \Omega, \mathcal{O}, \gamma)$, where $\mathcal{S}$ denotes the global state space.

$$\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N \quad (5)$$

where $\mathcal{A}$ denotes the joint action space, and $\mathcal{A}_i$ represents the action space of agent i . $\mathcal{P}(s'|s, \vec{a})$ denotes the state transition probability, describing the probability of transitioning from state s to s' under the joint action $\vec{a}$. $\mathcal{R}(s, \vec{a})$ is the global reward function.

$$\Omega = \Omega_1 \times \dots \times \Omega_N \quad (6)$$

where Ω is the joint observation space. $\mathcal{O}(\vec{o} | s, \vec{a})$ is the observation function, specifying the probability of obtaining the joint observation $\vec{o}$ after executing action $\vec{a}$ in state s . $\gamma \in [0, 1)$ is the discount factor.

In DC-STGN-MARL, each agent i receives a local observation $o_t^i \in \Omega_i$ at time t . This observation comprises its local inventory level I_t^i , in-transit inventory T_t^i , and the demand forecast $\hat{d}_{t+1:t+\tau}^i$ for the next τ periods along with its uncertainty estimate σ_t^i provided by the forecasting module:

$$o_t^i = (I_t^i, T_t^i, \hat{d}_{t+1:t+\tau}^i, \sigma_t^i) \quad (7)$$

Each agent selects an action a_t^i (typically an order quantity) based on its local policy $\pi_i(a_t^i | o_t^i)$. The actions of all agents collectively form the joint action $\vec{a}_t = (a_t^1, \dots, a_t^N)$. The system then transitions to a new state s_{t+1} according to the state transition function $\mathcal{P}$ and generates a global reward $r_t = \mathcal{R}(s_t, \vec{a}_t)$. This refers to the total quantity of goods ordered but not yet received, typically categorised by their expected arrival time within a defined lead time window. This reward function is designed as the total cost, specifically defined as:

$$r_t = - \left(\sum_{i=1}^N \left(c_h^i \cdot I_t^i + c_s^i \cdot \max(0, -I_t^i) + c_o^i \cdot \partial_{a_t^i > 0} \right) \right) \quad (8)$$

where c_h^i is the unit holding cost for agent i , c_s^i is the unit stockout cost, c_o^i is the fixed ordering cost, and $\partial_{a_t^i > 0}$ is the indicator function, which takes the value 1 when an order action occurs. The system's objective is to learn a set of cooperative policies:

$$\vec{\pi} = (\pi_1, \dots, \pi_N) \quad (9)$$

Maximise the expected cumulative discount reward

$$\mathbb{E}_{\vec{\pi}} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \right] \text{ (Liu et al., 2025).}$$

3.2 Deep spatiotemporal demand forecasting module

This module aims to provide the decision-making module with precise, fine-grained perception of future demand. As shown in Figure 1, this study models the dynamic management of e-commerce inventory as a decentralised partially observable Markov decision process (Dec-POMDP). propose a decomposition-coordinated spatiotemporal graph network (DC-STGN) to simultaneously capture the temporal dynamics of demand sequences and the spatial dependencies between commodities and warehouses.

First, this study constructed the entire inventory system as a graph structure:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \quad (10)$$

where node $v_i \in \mathcal{V}$ represents an inventory unit (e.g., a specific product in a specific warehouse), while each edge $e_{ij} \in \mathcal{E}$ represents an association between nodes (e.g., belonging to the same product category or the same geographic region). The edges can be binary, indicating mere existence of a relationship, or weighted, for instance, by the historical correlation coefficient of demand between the two nodes to represent the strength of their dependency. Each node v_i possesses a feature vector $\mathbf{x}_t^i$ at time t , containing historical demand sequences, prices, promotion flags, and other attributes.

The network adopts an encoder-decoder architecture. The encoder consists of stacked spatiotemporal convolutional blocks, each containing a temporal gated convolutional layer (TGC) and a graph attention network layer (GAT). Specifically, each node in $\mathcal{V}$ corresponds to a unique product-warehouse pair (e.g., 'Product A at Warehouse X'). Edges are established based on two types of associations:

- 1 product hierarchy (nodes belong to the same product category)
- 2 geographic proximity (warehouses serve overlapping regions).

These edges can be binary or weighted by historical demand correlation. The TGC layer captures temporal patterns specific to each node, with its operation expressed as:

$$\mathbf{Z}^{(l)} = \Phi(\mathbf{X}^{(l-1)} * \mathbf{\Gamma} + \mathbf{b}) \quad (11)$$

where $\mathbf{X}^{(l-1)}$ denotes the input to layer $l-1$, $*$ represents a one-dimensional causal convolution operation, $\mathbf{\Gamma}$ and $\mathbf{b}$ are the convolution kernel parameters and bias, respectively, and Φ is the GLU gated activation function (Lim and Zohren, 2021). The GAT layer is responsible for aggregating information from neighbouring nodes to update node representations. The DC-STGN encoder uses 4 stacked blocks, with GAT layers employing 4 attention heads and a hidden dimension of 128. The temporal convolutional kernel size is set to 7. For node v_i , its attention-based update is defined as:

$$\mathbf{h}_i^{(l)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}^{(l)} \mathbf{h}_j^{(l-1)} \right) \quad (12)$$

where $\mathcal{N}(i)$ denotes the set of neighbours of node v_i , $\mathbf{W}^{(l)}$ is the learnable weight matrix, and the attention coefficients are:

$$\alpha_{ij} = \text{softmax}_j \left(\text{LeakyReLU} \left(\mathbf{a}^T [\mathbf{W} \mathbf{h}_i \| \mathbf{W} \mathbf{h}_j] \right) \right) \quad (13)$$

where $\mathbf{a}$ is the attention vector. Although model interpretability is not the central focus of this work, the resulting attention weights α_{ij} can be post-hoc analysed to quantify the influence between nodes, potentially revealing non-obvious demand spillover or substitution effects between products or regions.

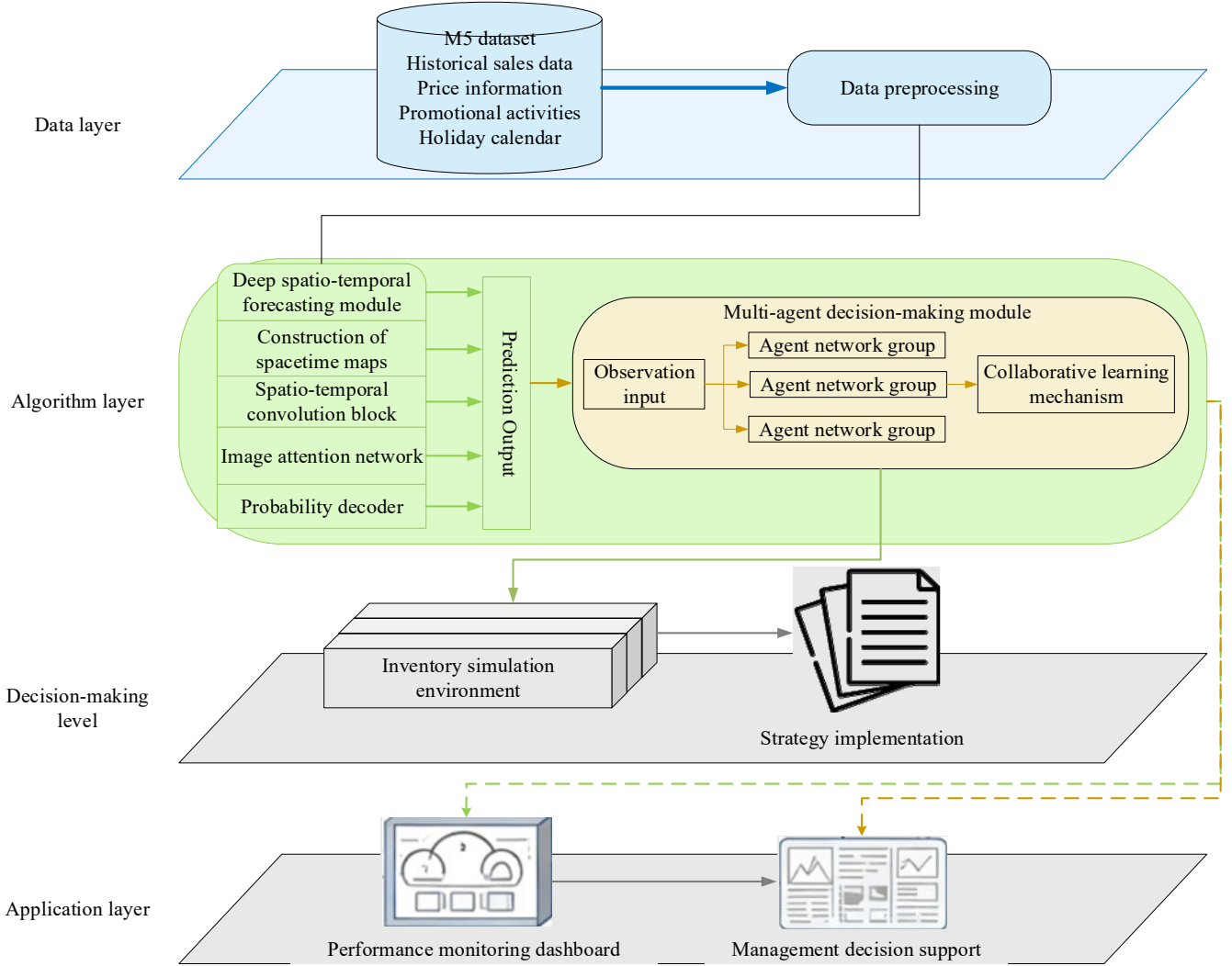
The decoder receives the context vector output from the encoder and generates a predictive distribution for the next τ time steps based on an autoregressive mechanism. This

study assume future demand follows a negative binomial distribution to better capture the highly discrete nature of count-level demand. The decoder outputs the parameters of this distribution: the mean $\hat{\mu}_{t+k}^i$ and dispersion $\hat{\phi}^i$. The final predicted value is $\hat{d}_{t+k}^i \sim \text{NegBinomial}(\hat{\mu}_{t+k}^i, \hat{\phi}^i)$. The loss function for this module is the negative log-likelihood:

$$\mathcal{L}_{\text{pred}} = - \sum_{i=1}^N \sum_{k=1}^{\tau} \log P(d_{t+k}^i | \hat{\mu}_{t+k}^i, \hat{\phi}^i) \quad (14)$$

In this manner, the prediction module not only provides a point estimate $\hat{\mu}$ but also outputs a quantified estimate σ of prediction uncertainty, which is crucial for subsequent risk-aware decision-making (Zhao et al., 2021). The forecast module was trained using the Adam optimiser with a learning rate of 0.001 and a batch size of 64 for 200 epochs.

Figure 1 Integrated simulation framework architecture diagram (see online version for colours)



3.3 Multi-agent collaborative decision-making module

The two modules are trained sequentially. First, the DC-STGN is pre-trained. Then, its output – the demand forecast mean and uncertainty (σ_t) – is concatenated with the inventory state to form the input feature vector for each agent's Q-network, which is then trained with the MARL algorithm. The decision module receives the output from the prediction module and uses it as partial observation to learn collaborative inventory strategies. This study adopts the VDN framework but incorporate key modifications to adapt it to the specific structure of inventory management.

The core idea of VDN is to decompose the joint action-value function $Q_{tot}(\vec{s}, \vec{a})$ of the system into the sum of individual agent action-value functions $Q_i(o_i, a_i; \theta_i)$:

$$Q_{tot}(\vec{s}, \vec{a}) \approx \sum_{i=1}^N Q_i(o_i, a_i; \theta_i) \quad (15)$$

where θ_i denotes the parameters of each agent's neural network. This decomposition enables centralised learning based on global rewards during training, while allowing each agent to independently select actions based on local Q_i values during execution [i.e., $a_i = \arg \max_{a_i} Q_i(o_i, a_i)$], thereby achieving scalable distributed decision-making (El Shar et al., 2022).

However, the standard VDN linear decomposition is overly simplistic to capture the complex nonlinear interactions between agents. Therefore, this study introduces a nonlinear hybrid network that takes individual Q_i values as input and fuses additional global state information s (such as global inventory levels and aggregate demand trends) to output a richer joint Q value estimate:

$$Q_{tot}(\vec{s}, \vec{a}) = g_\psi(s, Q_1(o_1, a_1), \dots, Q_N(o_N, a_N)) \quad (16)$$

where g_ψ is a nonlinear function parameterised by ψ , typically implemented using multilayer perceptron. This structure generalises the Q-value mixing network (QMIX) framework by imposing the monotonicity constraint $\frac{\partial g_\psi}{\partial Q_i} \geq 0$, thereby ensuring consistency between individual

optimal actions and the collective optimal action (Liu et al., 2025). Each agent's Q-network is a 2-layer multilayer perceptron (MLP) with 256 units per layer and ReLU activation.

To stabilise training, this study employs dual-Q learning combined with experience replay and periodic target network updates. The optimisation objective for the decision module is to minimise the temporal difference error:

$$\mathcal{L}_{\text{decision}}(\theta, \psi) = \mathbb{E}_{\mathcal{D}} \left[\left(r + \gamma \max_{\vec{a}'} Q_{tot}(\vec{s}', \vec{a}'; \theta^-, \psi^-) - Q_{tot}(\vec{s}, \vec{a}; \theta, \psi) \right)^2 \right] \quad (17)$$

where (θ^-, ψ^-) denotes the parameters of the target network, and $\mathcal{D}$ represents the empirical replay buffer, which stores transition samples $(s, \vec{a}, r, s')$ from system interactions. Training Details: The MARL module was trained using the Adam optimiser with a learning rate of 0.001. We employed an ϵ -greedy exploration strategy where ϵ decayed linearly from 1.0 to 0.05 over the course of training. The experience replay buffer had a capacity of 10^5 transitions, and the target network was updated every 100 steps. The reward was defined as the negative of the total cost [equation (7)], and training proceeded for 5×10^5 steps. The MARL module was trained using the Adam optimiser with a learning rate of 0.0005, a replay buffer size of 100,000, and a discount factor (γ) of 0.95. Training proceeded for 500,000 steps.

3.4 Simulation environment construction

The simulation process follows a standard inventory control loop: demand realisation $\rightarrow$ observation (inventory, forecast) $\rightarrow$ agent order decision $\rightarrow$ lead time delay $\rightarrow$ inventory update and cost calculation. To train and evaluate our integrated framework under conditions approximating real-world scenarios, this study constructed a high-fidelity simulation environment based on the M5 Forecasting dataset. This environment simulates a multi-warehouse, multi-product e-commerce inventory system, whose core components include:

- Demand generator: Generates daily granular demand aligned with real statistical characteristics based on actual historical sales data, calendar information (holidays), and promotional data provided in the M5 dataset. While anchored to historical patterns, the generator incorporates a stochastic component by sampling from the negative binomial distribution parameterised by our forecast module's output, thus explicitly simulating demand uncertainty.
- Inventory dynamics simulator: Updates inventory level I_t^i based on agent ordering actions a_t^i and random lead time L^i , which follows a LogNormal distribution $L \sim \text{LogNormal}(\mu = 3, \sigma = 1)$, while calculating holding costs and stockout costs.
- Reward calculator: Computes global rewards in real-time according to formula r_t . We performed sanity checks, such as verifying inventory level non-negativity, to ensure simulation correctness.

This environment provides an initial state for each training cycle, executes the agents' joint actions, and returns the next state and reward, forming a complete RL interaction loop. This approach enables large-scale, repeatable simulation experiments while preserving the complexity of real-world data (Gijbrenchts et al., 2022). The simulations and all machine learning models in this study were implemented using Python 3.9 and PyTorch 1.12 frameworks, and the random seeds were fixed in all experiments to ensure the reproducibility of the results.

4 Experimental validation

4.1 Experimental setup

To comprehensively evaluate the performance of the proposed integrated framework (hereafter referred to as DC-STGN-MARL), this study constructed a simulation environment based on the M5 Forecasting dataset (Makridakis et al., 2022). This dataset contains daily-granularity sales records for 3,049 products at Walmart over a 10-year period, along with rich covariate information such as price, promotional activities, and holiday calendars. This study selected the last 365 days as the test set, with the preceding data used for training and validation. The simulation environment replicates an e-commerce inventory network comprising three regional warehouses and one central warehouse. Each warehouse manages distribution for approximately 1,000 SKUs, with realistic cost parameters applied: unit holding cost $c_h = 0.02/\text{day}$, unit stockout cost $c_s = 1.5$, and fixed order cost $c_o = 0.5$. Unmet demand is treated as lost sales. The lost sales assumption was selected to model a competitive e-commerce environment where customers are unlikely to wait and will immediately turn to alternative sellers, which is a common scenario in online retail. All models were trained and tested under identical conditions to ensure fair comparison. The training and evaluation were conducted on a high-performance computing cluster equipped with multiple NVIDIA GPUs to manage the substantial computational load of the proposed DC-STGN-MARL framework and its baseline counterparts.

Simulation implementation and computational performance: Our simulation environment and models were implemented in Python 3.9 using PyTorch 1.12, and run on a server with an NVIDIA V100 GPU (32GB memory) and 128GB RAM. The training of the full DC-STGN-MARL framework required approximately 36 hours. The computational complexity of the MARL component scales linearly with the number of agents (warehouses) and the graph size. For model verification, we conducted sanity checks ensuring inventory conservation and compared simulated baseline policy outcomes against known analytical results where applicable. The implementation uses Python 3.9 with PyTorch 1.12. All experiments were run with a fixed random seed (42) to ensure reproducibility. Key hyperparameters are summarised in the revised methodological sections.

4.2 Baseline model and evaluation metrics

To comprehensively evaluate the performance of the proposed method, this study selected five representative baseline models for comparison, all sourced from authoritative literature in the relevant fields: Statistical baseline model:

- Seasonally adjusted autoregressive integrated moving average (SARIMA), with parameters determined via the AIC criterion (Lewis-Beck, 2005).

- Deep learning forecasting models: Frequency enhanced decomposed transformer (FEDformer) (Zhou et al., 2022), a frequency-enhanced transformer model demonstrating exceptional performance in long-term sequence forecasting; and neural basis expansion analysis for time series forecasting (N-BEATS) (Oreshkin et al., 2019), a pure deep learning architecture achieving outstanding results in the M4 competition.
- Prediction-based optimisation strategy: Incorporating FEDformer predictions into an optimised (s, S) inventory policy for decision-making.
- DRL models: Single-agent DRL (Gijssbrechts et al., 2022), treating the entire system as a centrally controlled agent; and Independent DQN, where each warehouse learns as an independent DQN agent without explicit coordination (Tan, 1993).

This study employ four key performance indicators (KPIs) for evaluation, while calculating their 95% confidence intervals: Forecast Accuracy: WMAPE, defined as

$$WMAPE = \frac{\sum |y_t - \hat{y}_t|}{\sum y_t},$$

where y_t is the actual value and $\hat{y}_t$ is the forecast value.

- Operating costs: Average total cost per unit time, including carrying costs, stockout costs, and ordering costs.
- Service quality: Measured by the order fill rate, defined as

$$Fill\ Rate = 1 - \frac{Total\ Shortfalls}{Total\ Demand}.$$

- Asset efficiency: Inventory turnover, calculated as $\frac{Cost\ of\ Goods\ Sold}{Average\ Inventory\ Value}$.

To validate the statistical significance of performance improvements, this study conducted 30 independent replicate experiments across all models and performed paired t-tests on key KPIs, calculating p-values and Cohen's d effect sizes.

In addition, to gain a deeper understanding of the sources of performance differences, we further analysed the composition of total costs (including changes in the respective proportions of holding costs, backorder costs, and order costs), the stability of service levels (measured by the variance of order fulfilment rates over the test period), and the drivers of improvement in inventory turnover (e.g., due to a general reduction in safety stocks, Or the improvement in the prediction accuracy of a specific node, or the effect of a collaborative replenishment strategy).

4.3 Results analysis

Quantitative results show that our proposed DC-STGN-MARL framework significantly outperforms all baseline models across most metrics. As shown in Figure 2, in terms of prediction accuracy, DC-STGN-MARL achieves a WMAPE of $6.35\% \pm 0.18\%$. Compared to the second-best model, FEDformer ($7.02\% \pm 0.21\%$), this improvement is highly statistically significant ($p < 0.01$, Cohen's $d = 0.82$). This confirms the effectiveness of the proposed DC-STGN in capturing complex demand patterns. Notably, our model demonstrates a more pronounced predictive advantage during periods of intense demand volatility, such as promotional periods and holidays, as illustrated by the time-series prediction comparison plot in Figure 3. This indicates that its modelling of spatiotemporal dependencies better accommodates sudden demand shifts.

In core inventory management performance, DC-STGN-MARL demonstrates a more pronounced advantage. As clearly shown in Figure 4, our model reduces the average total cost per unit time to 125.4 ± 3.7 , representing reductions of 13.6% and 9.5% compared to the single-agent DRL (145.2 ± 4.1) and FEDformer+(s,S) strategy (138.6 ± 3.9), respectively. Both differences are highly statistically significant ($p < 0.01$). This outcome underscores the value of end-to-end learning that integrates precise forecasting with collaborative decision-making. Analysis of the cost structure reveals that DC-STGN-MARL significantly reduces safety stock levels while maintaining comparable service levels through more accurate forecasting and collaborative replenishment. This optimises the trade-off between holding costs and stockout costs.

In terms of service quality, all models achieved an order fulfilment rate exceeding 90%. However, DC-STGN-MARL ranked first with a performance of

$95.8\% \pm 0.5\%$ and exhibited the smallest fluctuation range, indicating its strategy is the most robust. Crucially, in asset efficiency, DC-STGN-MARL achieved an inventory turnover rate of 8.4 ± 0.3 , significantly higher than FEDformer+(s, S)'s 7.1 ± 0.2 ($p < 0.01$). This demonstrates DC-STGN-MARL maintains high service levels while accelerating inventory turnover and enhancing capital utilisation efficiency. The primary driver is the framework's ability to synchronise replenishment decisions across the network, reducing redundant safety stock and aligning inventory levels more closely with spatiotemporally aware demand forecasts.

Figure 2 Comparison of demand forecasting accuracy across models (see online version for colours)

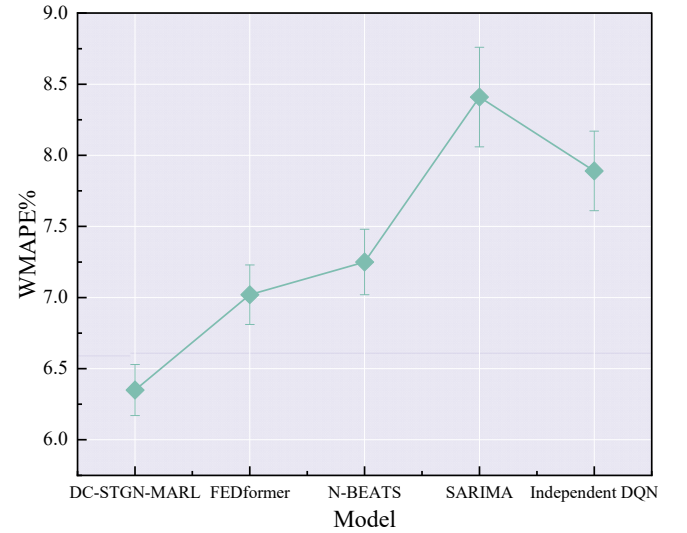


Figure 3 Representative product demand forecast time series comparison (see online version for colours)

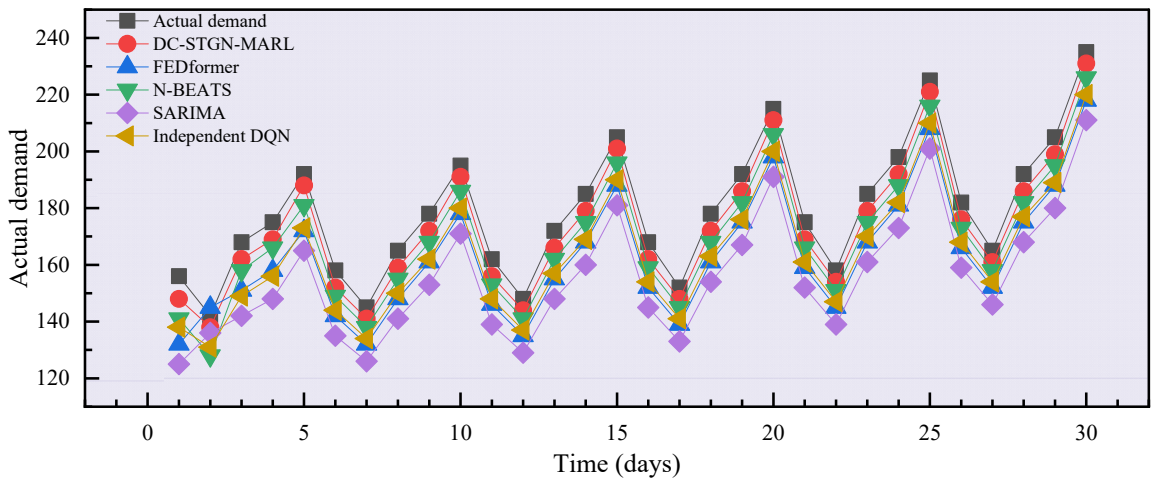


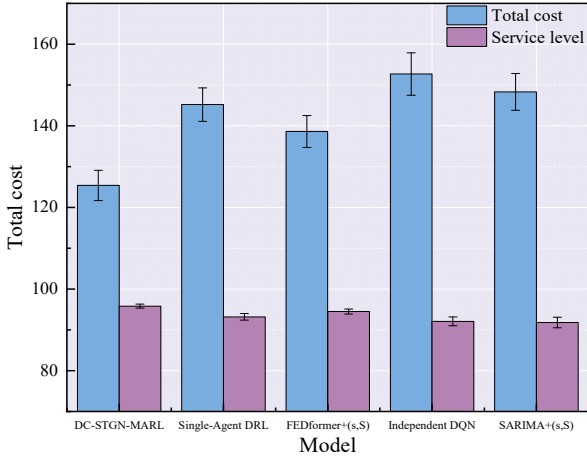
Figure 4 Comparison of total costs and service levels across models (see online version for colours)

Table 1 details ablation experiment results, further validating the necessity of each framework component. When removing the GNN component and using only the temporal model (DC-STGN-MARL w/o GNN), WMAPE increased to 6.91% and total cost rose to 135.2. This demonstrates that ignoring spatial correlations between goods degrades prediction accuracy and leads to suboptimal decisions. When replacing the multi-agent coordination mechanism with independent learning (DC-STGN-MARL w/o coordination), the total cost surged to 152.7 while service level dropped to 92.1%. This strongly demonstrates that explicitly modelling coordination among agents is crucial for achieving global optimisation in distributed inventory systems. Comparing the variants, the performance degradation is most acute when the collaborative mechanism is removed, indicating that effective inter-agent coordination is even more critical for overall cost reduction than capturing fine-grained spatial dependencies in this specific setting.

Table 1 Analysis of ablation study results

Model variants	WMAPE (%)	Total cost (\$)	Service level (%)	Inventory turnover ratio
Full model	6.35	125.4	95.8	8.4
No GNN components	6.91	135.2	94.2	7.6
No collaborative mechanism	7.25	152.7	92.1	6.9
Time-series model only	7.89	145.8	93.5	7.1

In order to highlight the advantages of DC-STGN-MARL (DC-STGN-MARL), we emphatically compare it with three representative methods. Firstly, compared with traditional static strategies (e.g., (s, S) strategy), DC-STGN-MARL shows stronger adaptability in the face of complex spatiotemporal demand fluctuations in M5 data, achieving 13.6% total cost reduction. Secondly, compared with Independent DQN, our coordination mechanism significantly improves the overall service level of the system (the order fulfilment rate increases by 3.7 percentage

points), which proves the necessity of multi-agent cooperation. Third, compared with the strategy of coupling optimised decision with advanced prediction-only models such as FEDformer, our end-to-end framework achieves a further optimisation of 0.67% on WMAPE by directly incorporating predictive uncertainty into the decision-making process, demonstrating the value of tight integration of forecasting and decision making.

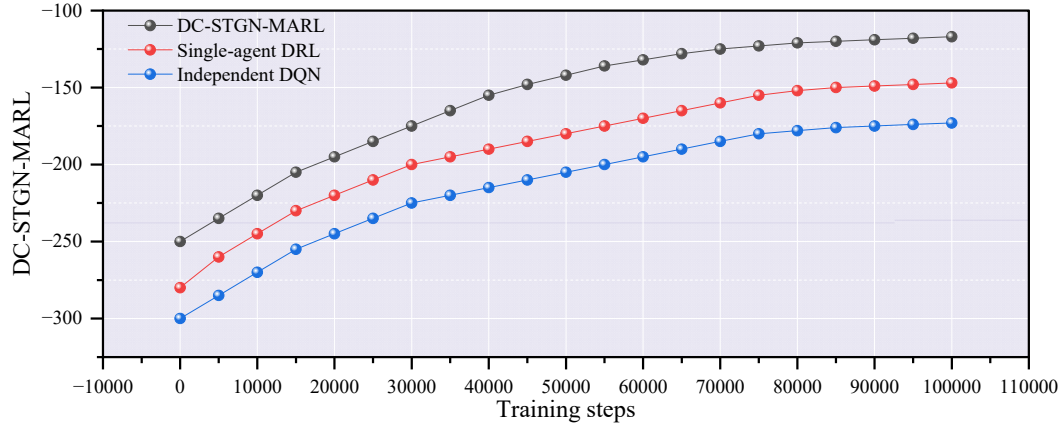
We further conducted sensitivity analyses on key hyperparameters. Varying the discount factor γ within $\{0.9, 0.95, 0.99\}$ showed that $\gamma = 0.95$ yielded the most stable and cost-effective policy. Altering the forecasting horizon τ among $\{7, 14, 30\}$ days indicated that $\tau = 14$ provided the best balance between forecast accuracy and operational cost for our setting. These results demonstrate the robustness of DC-STGN-MARL within reasonable parameter ranges.

4.4 Computational performance analysis

The training of the full DC-STGN-MARL framework was conducted on a single NVIDIA V100 GPU (32GB VRAM) and required approximately 36 hours in total. This decomposes into 12 hours for pre-training the DC-STGN forecast module and 24 hours for training the MARL policy. As illustrated by the convergence curves in Figure 5, DC-STGN-MARL not only achieves a lower final cost but also converges approximately 25% faster (reaching a stable policy within 100k episodes) compared to the Single-Agent DRL and Independent DQN baselines. This acceleration in learning stems from the more informative and stable state representation provided by the integrated forecast module. In terms of scalability, the per-step computational complexity of the MARL component is approximately linear in the number of agents N and graph nodes $|V|$, i.e., $O(N|V|)$. While this enables handling of networks like the one in our M5-based experiment ($|V| \approx 4,000$), deployment to significantly larger networks (e.g., $>10,000$ nodes) would benefit from distributed training strategies, which is a direction for future work.

4.5 Experimental results and analysis

The experimental results comprehensively validate the effectiveness of the proposed integrated framework. The success of DC-STGN-MARL can be attributed to two core design elements: first, it simultaneously leverages temporal patterns and spatial correlations in demand data through a deep spatiotemporal graph network, enabling more precise and robust demand sensing; second, it tightly integrates predictive information with the global objective of inventory optimisation through a multi-agent collaborative decision-making mechanism during end-to-end training, achieving a unified approach between local decisions and global optimisation. This aligns with recent research emphasising the fusion of forecasting and decision-making (Zhao et al., 2021) and offers a feasible solution for addressing large-scale distributed inventory coordination optimisation.

Figure 5 Comparison of model training convergence curves (see online version for colours)

As shown in Figure 5, our method demonstrates superior convergence speed and stability compared to baseline DRL models, owing to its more efficient state representation and collaborative learning mechanism. However, this study has several limitations. First, although the simulation environment is based on real data, assumptions such as infinite supplier capacity and fixed lead times may oversimplify some real-world constraints. Future work could incorporate more complex supply chain disruptions and uncertainties. Second, the current model incurs significant computational overhead during training. Improving training efficiency to enable faster policy iteration is a direction worth exploring.

Looking ahead, this research can be further deepened in several directions: First, incorporating external environmental factors (such as macroeconomic indicators and market competition dynamics) into the model framework could enhance the system's adaptability to external shocks (Guo and Mantravadi, 2025). Second, exploring more interpretable artificial intelligence methods could make the model's decision-making process more transparent, facilitating managerial understanding and acceptance. Finally, with the advancement of emerging technologies like quantum computing, research can investigate how these technologies can address computational challenges in solving hyper-scale inventory optimisation problems. This study believes that as artificial intelligence further integrates with operations management, intelligent inventory management will play an increasingly vital role in enhancing supply chain resilience and driving sustainable development.

5 Discussion

Through systematic experimental validation, this study reveals the significant advantages of the integrated framework combining deep spatiotemporal networks with MARL in e-commerce inventory management. Experimental results consistently demonstrate that our proposed DC-STGN-MARL model significantly

outperforms traditional methods and existing state-of-the-art models across multiple dimensions, including prediction accuracy, operational costs, and service levels. This success not only validates the technical solution's effectiveness but, more importantly, highlights its underlying theoretical value and practical significance. From a theoretical perspective, the core contribution lies in successfully bridging prediction and decision-making, effectively resolving the long-standing research divide between operations research and artificial intelligence (Zhao et al., 2021). Traditional inventory management research often treats forecasting and decision-making as two independent stages: forecasting models pursue statistical accuracy, while decision models optimise based on given forecasts. This fragmented approach overlooks the substantial impact of forecast error distributions on decision outcomes. Our end-to-end framework achieves a leap from 'accurate forecasting' to 'optimal decision-making' by directly integrating forecast uncertainty into the decision process.

Of particular note, the superior performance demonstrated by the multi-agent coordination mechanism in this study offers novel insights for optimising distributed inventory systems. Compared to single-agent models, our approach not only considers the local information of each warehouse but also captures the synergistic effects within the system through a Collaborative decision network. This aligns closely with the global optimisation philosophy emphasised in modern supply chain management (Liu et al., 2025). Experimental results demonstrate that incorporating the coordination mechanism reduces total system costs by 13.6%. This improvement is not only statistically significant but also holds substantial practical value. Furthermore, the success of spatiotemporal graph networks in capturing complex inter-product relationships expands our understanding of the spatial dimension's importance in demand forecasting. Traditional forecasting models often treat demand sequences for individual commodities in isolation. Our approach, however, explicitly models substitution and complementarity relationships between commodities through graph attention mechanisms. This partially explains why our model performs particularly well

in scenarios with volatile demand, such as promotional periods and holidays (Xu et al., 2022).

However, this study must also clearly recognise the limitations of this study. Although the simulation environment is constructed based on real data, simplified assumptions regarding supplier capacity and lead times may diverge from real-world business conditions. Future research should consider incorporating more complex supply chain network structures and integrating additional real-world uncertainties, such as supplier reliability and transportation disruption risks (Shishodia et al., 2023). Additionally, the current model exhibits high computational complexity, potentially posing efficiency challenges when applied to hyperscale inventory systems. This highlights the need to strike a better balance between algorithmic efficiency and decision quality. The modelling in this study is based on the following key assumptions:

- 1 The order lead time $L_{sim} \sim \text{LogNormal}(\mu = 3, \sigma = 1)$ follows a lognormal distribution.
- 2 An unmet customer demand is considered a permanent loss.
- 3 The supplier's supply capacity is unlimited.
- 4 In the simulated environment, daily demand follows a negative binomial distribution parameterised by the prediction module.

These assumptions simplify the problem and focus on the core research points, but also constitute the limitations of the model. Firstly, the performance of the model is highly dependent on the quality and representation of historical data. Secondly, the training of the framework requires considerable GPU computing resources and time. Finally, current models do not incorporate external uncertainties such as supply chain disruptions, dynamic pricing, or intense market competition. Future research will focus on expanding in these directions to enhance the robustness and practicability of the model.

Looking ahead, this research can be further deepened in several directions: first, incorporating external environmental factors (such as macroeconomic indicators and market competition dynamics) into the model framework could enhance the system's resilience to external shocks (Guo and Mantravadi, 2025); Second, more interpretable AI methods could be explored to make the model's decision-making process more transparent, facilitating managerial understanding and acceptance. Finally, with the advancement of emerging technologies like quantum computing, research could investigate how these technologies can address computational challenges in hyperscale inventory optimisation. This study believes that as AI and operations management continue to converge, intelligent inventory management will play an increasingly vital role in enhancing supply chain resilience and promoting sustainable development.

6 Conclusions

This study addresses the challenge of dynamic inventory management in e-commerce environments by proposing an integrated simulation framework that combines deep spatiotemporal networks with MARL. Through systematic theoretical analysis and empirical validation, we draw the following key conclusions: First, the end-to-end joint optimisation paradigm for forecasting and decision-making significantly outperforms traditional two-stage approaches, with statistically significant validation across multiple key performance indicators; Second, in multi-warehouse collaborative scenarios, the value decomposition method considering agent interactions effectively enhances overall system performance, preventing suboptimal decisions caused by local optimisation. Finally, explicitly modelling inter-item relationships through spatiotemporal graph networks significantly improves demand forecasting accuracy, particularly in scenarios with high demand volatility.

The theoretical contributions of this study manifest in three dimensions: First, this study bridge the theoretical gap between forecasting and decision models by establishing a unified theoretical framework. Second, this study introduces multi-agent collaborative learning into inventory management, expanding the methodological boundaries of distributed decision optimisation. Third, this study proposes a novel spatiotemporal demand representation method, enriching the theoretical toolkit for demand forecasting. Practically, this research provides actionable intelligent inventory management solutions for e-commerce enterprises. Companies can leverage the proposed framework to build their own digital twin systems, using simulation to evaluate and optimise inventory strategies, thereby reducing operational costs and enhancing service levels.

Acknowledgements

This work is supported by the 2022 Teacher 'Innovation' Project of Harbin University of Commerce titled "The Study of Mechanisms and Strategies on the Role of Heilongjiang Local Government in Promoting Agriculture through Digital Commerce" (No. XW0132).

Declarations

All authors declare that they have no conflicts of interest.

References

- Castro Moraes, T., Qin, J., Yuan, X-M. and Chew, E.P. (2023) 'Evolving hybrid deep neural network models for end-to-end inventory ordering decisions', *Logistics*, Vol. 7, No. 4, p.79.
- Chong, J.W., Kim, W. and Hong, J. (2022) 'Optimization of apparel supply chain using deep reinforcement learning', *IEEE Access*, Vol. 10, No. 1, pp.100367–100375.

- Dey, B.K. and Seok, H. (2024) 'Intelligent inventory management with automation and service strategy', *Journal of Intelligent Manufacturing*, Vol. 35, No. 1, pp.307–330.
- El Shar, I., Sun, W., Wang, H. and Gupta, C. (2022) 'Deep reinforcement learning toward robust multi-echelon supply chain inventory optimization', *Automation Science and Engineering*, Vol. 10, No. 1, pp.1385–1391.
- Gijsbrechts, J., Boute, R.N., Van Mieghem, J.A. and Zhang, D.J. (2022) 'Can deep reinforcement learning improve inventory management? Performance on lost sales, dual-sourcing, and multi-echelon problems', *Manufacturing & Service Operations Management*, Vol. 24, No. 3, pp. 1349–1368.
- Guo, D. and Mantravadi, S. (2025) 'The role of digital twins in lean supply chain management: review and research directions', *International Journal of Production Research*, Vol. 63, No. 5, pp.1851–1872.
- Harahap, A.Z.M.K., Rahim, M.K.I.A., Malinjasari, N., Salleh, S.M. and Ma'arof, R.A. (2025) 'Enhancing the inventory management through demand forecasting', *International Journal of Research and Innovation in Social Science*, Vol. 9, No. 1, pp.2737–2744.
- Jackson, I., Tolujevs, J. and Kegenbekov, Z. (2020) 'Review of inventory control models: a classification based on methods of obtaining optimal control parameters', *Transport and Telecommunication*, Vol. 21, No. 3, pp.191–202.
- Jing, Z. and Zhang, H. (2023) 'Intelligent traffic assignment method of urban traffic network based on deep reinforcement learning', *International Journal of Information and Communication Technology*, Vol. 22, No. 1, pp.60–72.
- Lewis-Beck, M.S. (2005) 'Election forecasting: Principles and practice', *The British Journal of Politics and International Relations*, Vol. 7, No. 2, pp.145–164.
- Lim, B. and Zohren, S. (2021) 'Time-series forecasting with deep learning: a survey', *Philosophical Transactions of the Royal Society A*, Vol. 379, No. 2194, p.20200209.
- Liu, X., Hu, M., Peng, Y. and Yang, Y. (2025) 'Multi-agent deep reinforcement learning for multi-echelon inventory management', *Production and Operations Management*, Vol. 34, No. 7, pp.1836–1856.
- Makridakis, S., Spiliotis, E. and Assimakopoulos, V. (2022) 'The M5 competition: Background, organization, and implementation', *International Journal of Forecasting*, Vol. 38, No. 4, pp.1325–1336.
- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K. and Ostrovski, G. (2015) 'Human-level control through deep reinforcement learning', *Nature*, Vol. 518, No. 7540, pp.529–533.
- Oreshkin, B.N., Carpow, D., Chapados, N. and Bengio, Y. (2019) 'N-BEATS: Neural basis expansion analysis for interpretable time series forecasting', *Machine Learning*, Vol. 97, No. 1, pp.4997–5006.
- Rishab, A. (2022) 'Deep reinforcement learning for inventory management', *Nanyang Technological University*, Vol. 1, No. 1, p.1.
- Shishodia, A., Sharma, R., Rajesh, R. and Munim, Z.H. (2023) 'Supply chain resilience: A review, conceptual framework and future research', *The International Journal of Logistics Management*, Vol. 34, No. 4, pp.879–908.
- Sobh, T.S. (2024) 'A secure and integrated ontology-based fusion using multi-agent system', *International Journal of Information and Communication Technology*, Vol. 25, No. 1, pp.48–73.
- Tan, M. (1993) 'Multi-agent reinforcement learning: independent vs. cooperative agents', *Machine Learning*, Vol. 1, No. 1, pp.330–337.
- Thrun, S. and Littman, M.L. (2000) 'Reinforcement learning: an introduction', *AI Magazine*, Vol. 21, No. 1, pp.103–103.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I. (2017) 'Attention is all you need', *Advances in Neural Information Processing Systems*, Vol. 30, No. 1, p.1.
- Wang, J., Chong, W.K., Lin, J. and Hedenstierna, C.P.T. (2024) 'Retail demand forecasting using spatial-temporal gradient boosting methods', *Journal of Computer Information Systems*, Vol. 64, No. 5, pp.652–664.
- Xu, C., Huang, H., Ying, X., Gao, J., Li, Z., Zhang, P., Xiao, J., Zhang, J. and Luo, J. (2022) 'HGNN: hierarchical graph neural network for predicting the classification of price-limit-hitting stocks', *Information Sciences*, Vol. 607, No. 1, pp.783–798.
- Zhao, C., Wan, C. and Song, Y. (2021) 'Cost-oriented prediction intervals: on bridging the gap between forecasting and decision', *IEEE Transactions on Power Systems*, Vol. 37, No. 4, pp.3048–3062.
- Zhou, T., Ma, Z., Wen, Q., Wang, X., Sun, L. and Jin, R. (2022) 'Fedformer: frequency enhanced decomposed transformer for long-term series forecasting', *Machine Learning*, Vol. 1, No. 1, pp.27268–27286.

List of abbreviations

DRL	Deep reinforcement learning
GNN	Graph neural network
LSTM	Long short-term memory
MARL	Multi-agent reinforcement learning
MDP	Markov decision process
RL	Reinforcement learning
VDN	Value decomposition networks
WMAPE	Weighted mean absolute percentage error.